Reverse Engineering Activity

Logitech M331 Silent Plus

Observe → Research → Tear down → Identify → Reason

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Electronic Product Design & Reverse Engineering (26EC106PCB)



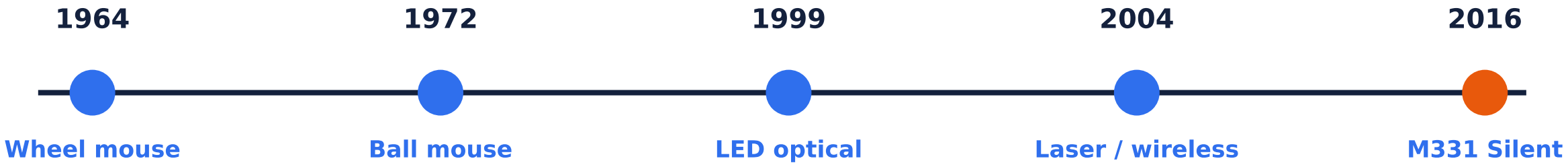
1 · Physical Inspection (before opening)

Observed	M331
Model / maker	M331 Silent Plus · Logitech (2016)
Size	~106 × 68 × 39 mm, right-handed
Inputs	L/R buttons, scroll wheel + click
Power	1 × AA, ON/OFF slide switch
Bottom	Optical window, feet, battery door
Connector	None — 2.4 GHz USB nano receiver
Seams / screws	No visible screws (hidden)

Three questions

- **Purpose:** move a cursor, click, scroll — quietly.
- **User expects:** no cable, long battery life, comfort, silent clicks.
- **Hypothesis:** AA → regulator → MCU + radio → USB receiver; window → optical sensor.

2 · Version History — and Why It Changed



Parameter	Old (ball mouse)	M331 (2016)	Why changed
Tracking	Rubber ball + rollers	Optical sensor, 1000 DPI	No dirt, more precise
Connection	Cable (PS/2, USB)	2.4 GHz wireless	Freedom of movement
Power	From the PC	1 × AA, ~18 months	Low-power chips
Clicks	Standard switch	>90% quieter	Shared/quiet spaces

Evidence: Logitech specs + teardown. Hypothesis: the 'why changed' column — our engineering reasoning.

3 · User Needs & Design Trade-offs

What users want

- Quiet clicks
- No cable, 10 m range
- Long battery life
- Comfortable grip
- Plug-and-play

What the designer had to balance (likely)

1 × AA battery

+ cheap, swappable, ~18 months
− not rechargeable, adds weight

1000 DPI sensor

+ enough for office, low power, low cost
− not for gaming precision

Silent switches

+ quiet in shared spaces
− softer click feedback

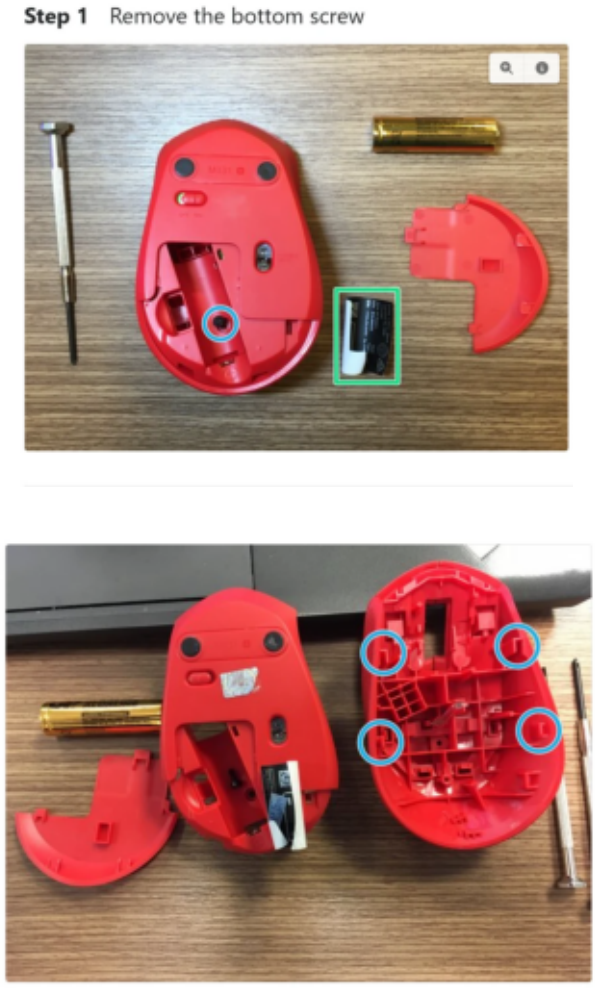
2 screws + snap fits

+ fast, cheap assembly
− harder to open / repair

4 · Guided Teardown & ESD Safety

Document → Photograph → Open → Photograph → Identify → Record

- **Safety first:** battery out; ground yourself; hold PCB by its edges; don't touch IC pins.
- **Tools:** precision screwdriver, plastic pry tool, tweezers, phone camera.
- **Hidden screw:** under the label in the battery well — the only case screw.
- **Shell:** slide forward; 4 snap hooks release.
- **PCB:** 1 screw + plastic pins; wheel lifts out.

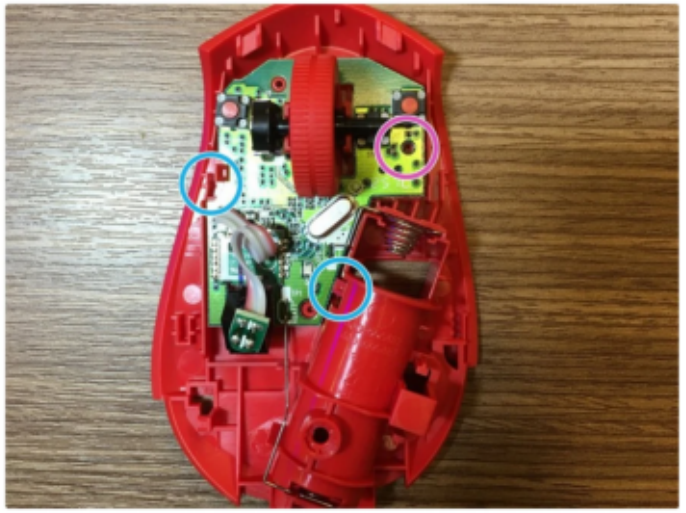


Hidden screw & snap-hook clips (photos: iFixit, CC BY-NC-SA)

5 · Inside the M331



All parts laid out (photos: iFixit, CC BY-NC-SA)



Main PCB in the base

- **Housing:** 4 plastic parts
- **Wheel:** + encoder
- **PCB:** switches, crystal, SoC, antenna
- **Optics:** clear lens
- **Fasteners:** 2 screws, spring

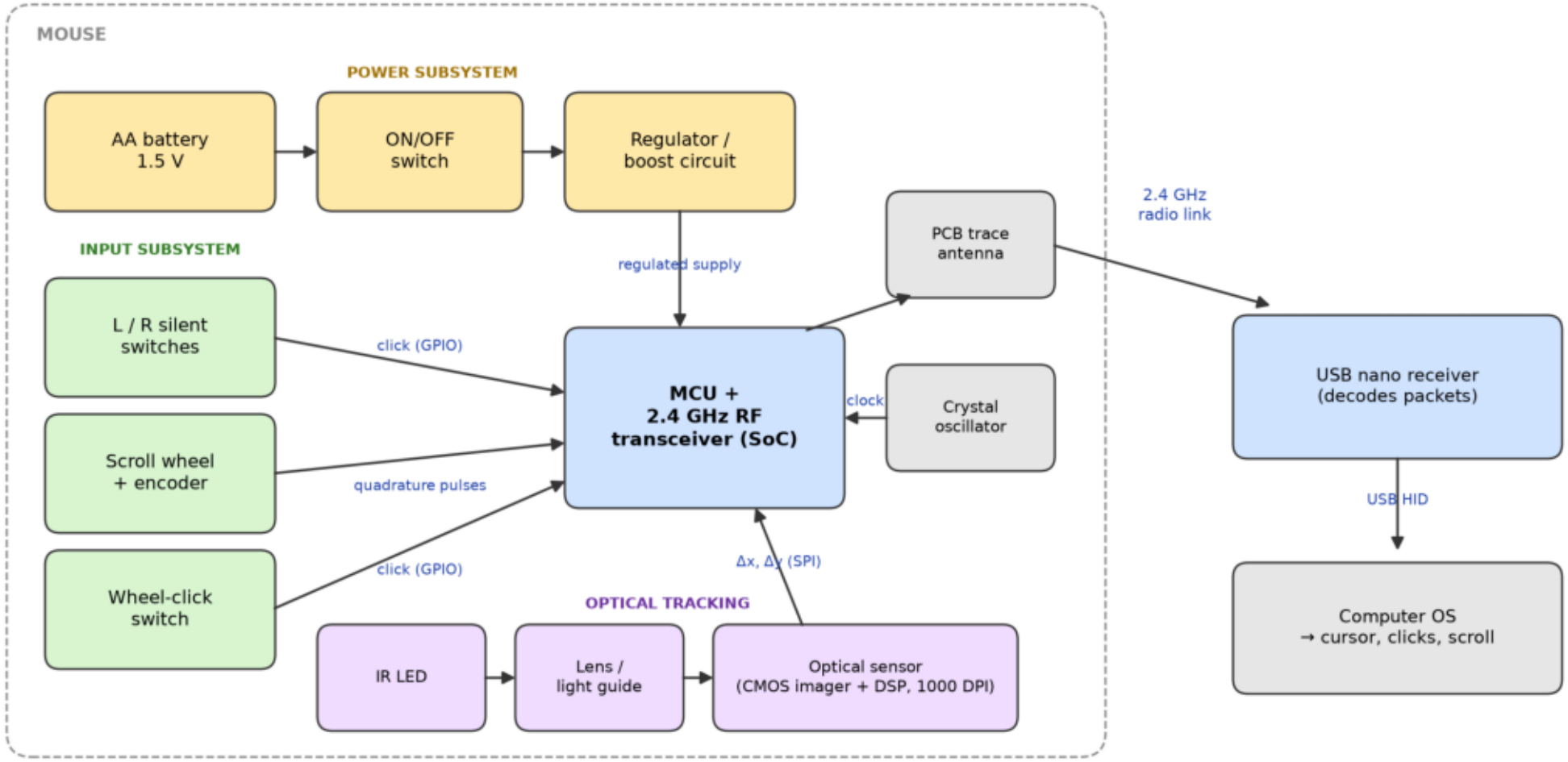
6 · Components & Their Functions

Component	Electrical role	Functional role	Connected to
MCU + 2.4 GHz SoC	Processing + radio	Product controller	All inputs, antenna
Optical sensor	CMOS imager + DSP	Measures movement	MCU (SPI)
IR LED + lens	Illumination / optics	Lights the desk	Sensor
Crystal oscillator	Clock source	Radio timing	SoC
Silent switches ×2	Contact closure	L / R click	MCU GPIO
Wheel encoder	Quadrature pulses	Scroll + direction	MCU GPIO
PCB trace antenna	RF radiator	Wireless link	SoC
AA + regulator	Power conversion	Supplies all chips	Battery → SoC

Next step: read the IC markings → datasheets → exact part numbers & ratings.

7 · System Architecture (Block Diagram)

Logitech M331 — Internal Functional Block Diagram



Schematic functional view; exact chip part numbers not published by Logitech — read IC markings on the PCB to complete.

8 · How It Works — Step by Step

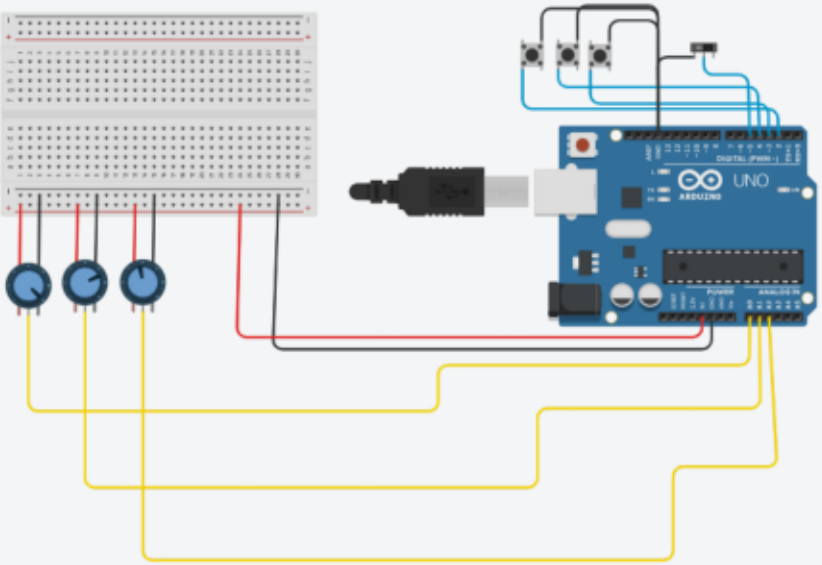


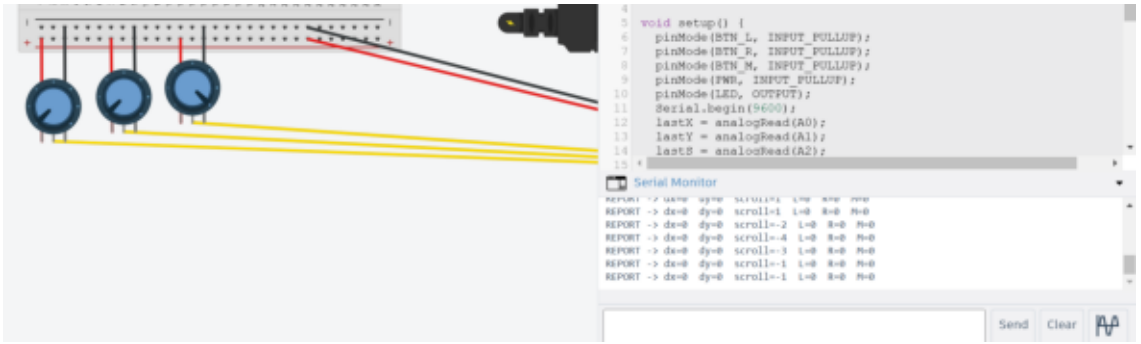
Clicks: switch closes → MCU debounces → bit set in the report.

Scroll: two offset pulse trains — their order gives the direction.

Idle: low-power sleep; any movement or click wakes it instantly.

9 · Verifying It: Tinkercad Simulation





- **Stand-ins:** 2 pots = sensor X/Y, 1 pot = scroll, 3 buttons = clicks, slide switch = power, Arduino = MCU.
- **Result:** switch ON lit the LED; knobs gave dx/dy/scroll; buttons gave L/R/M = 1; OFF stopped reports.

The simulation reproduces the input → processing → report flow of the real mouse.

10 · Five Levels of Thinking — Conclusions

1 Observation — AA cell, 3 buttons, optical window, no visible screws

2 Identification — MCU + 2.4 GHz SoC, optical sensor, crystal, silent switches

3 Specification — 1000 DPI · 2.4 GHz · 10 m · ~18 months on one AA

4 Function — Sensor + switches → MCU → radio → receiver → cursor

5 Reasoning — Designer chose low-power SoC + 1 AA for long life at low cost, silent switches for quiet spaces, snap fits for cheap assembly

Thank you · Questions?